

HOLBAEK, Denmark

WMO: 061560

Lat: 55.7358N

Lon: 11.6033E

Elev: 13

StdP: 101.17

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -8.0	(c) -5.9	(d) -10.6	(e) 1.5	(f) -6.2	(g) -8.7	(h) 1.8	(i) -4.6	(j) 20.1	(k) 5.9	(l) 15.7	(m) 6.2	(n) 2.4	(o) 70	(p) 0.643

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 9.4	(c) 27.0	(d) 19.0	(e) 25.2	(f) 18.3	(g) 23.5	(h) 17.6	(i) 20.2	(j) 24.7	(k) 19.2	(l) 23.5	(m) 18.4	(n) 22.2	(o) 4.9	(p) 110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 18.6	(b) 13.4	(c) 22.0	(d) 17.7	(e) 12.7	(f) 21.1	(g) 16.8	(h) 12.0	(i) 20.2	(j) 57.9	(k) 24.8	(l) 54.8	(m) 23.6	(n) 51.9	(o) 22.5	(p) 23.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
13.9	11.4	9.8	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			-12.3	29.4	5.0	1.8	-15.9	30.7	-18.8	31.7	-21.6	32.7	-25.3	34.0
			-12.5	21.8	4.8	0.9	-16.0	22.5	-18.8	23.0	-21.5	23.5	-25.0	24.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	9.0	0.9	1.2	3.4	7.8	12.0	14.8	17.7	17.5	14.1	9.7	5.8	2.6
	DBStd	6.76	3.83	3.55	3.49	3.15	3.16	2.57	2.75	2.50	2.48	3.10	3.05	3.89
	HDD10.0	1233	283	248	204	81	14	1	0	0	2	43	128	228
	HDD18.3	3461	542	481	462	316	197	110	47	47	128	268	376	487
	CDD10.0	867	0	0	1	14	77	144	238	233	124	33	3	0
	CDD18.3	54	0	0	0	0	1	3	26	22	2	0	0	0
	CDH23.3	392	0	0	0	1	9	38	231	108	6	0	0	0
	CDH26.7	56	0	0	0	0	1	2	39	14	0	0	0	0
Wind	WSAvg	4.7	5.4	5.3	5.1	4.7	4.3	4.4	3.8	4.0	4.2	4.7	5.0	5.1
Precipitation	PrecAvg	627	50	38	36	34	41	62	60	70	62	63	55	56
	PrecMax	784	88	81	78	68	81	130	140	152	170	132	108	102
	PrecMin	423	4	6	9	7	14	0	5	21	14	25	9	19
	PrecStd	94	23	16	19	16	18	33	31	37	37	24	24	22
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.7	9.9	14.1	20.1	24.7	26.4	29.6	28.6	24.2	19.1	13.7	11.0
		MCWB	8.7	7.9	9.6	13.3	17.8	18.5	20.2	19.9	17.8	15.5	12.6	9.9
	2%	DB	8.1	8.4	11.4	17.7	22.1	24.3	27.7	26.2	21.9	16.2	12.2	9.8
		MCWB	7.2	7.0	8.4	11.9	15.4	17.9	19.3	18.5	16.9	14.0	11.2	8.9
	5%	DB	7.0	7.2	9.4	15.4	20.0	22.3	26.1	24.4	19.9	15.0	11.1	8.5
		MCWB	6.1	6.0	7.1	10.5	14.7	16.8	18.9	18.0	15.9	13.3	10.1	7.7
	10%	DB	6.0	6.1	8.1	13.4	18.1	20.3	24.1	22.7	18.3	13.9	9.9	7.5
		MCWB	5.2	5.1	6.4	9.6	13.5	15.7	18.1	17.4	15.0	12.4	9.0	6.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.9	8.6	10.3	14.2	18.8	20.2	21.7	21.1	18.6	16.1	12.8	10.2
		MCDB	9.7	9.4	12.6	19.1	23.2	24.6	27.4	25.5	22.2	17.9	13.5	10.8
	2%	WB	7.4	7.2	8.6	12.5	16.7	18.7	20.5	19.8	17.3	14.5	11.4	9.0
		MCDB	8.0	8.1	10.7	16.7	20.3	22.9	25.5	23.8	20.5	15.8	12.0	9.7
	5%	WB	6.2	6.2	7.6	11.2	15.2	17.5	19.6	18.9	16.5	13.5	10.2	7.7
		MCDB	6.9	7.1	9.1	14.4	18.7	21.1	24.3	22.6	19.2	14.7	10.9	8.4
	10%	WB	5.2	5.1	6.5	9.9	14.0	16.2	18.7	18.1	15.6	12.6	9.2	6.8
		MCDB	5.9	6.0	7.9	12.8	17.5	19.5	22.9	21.7	17.8	13.7	9.8	7.4
Mean Daily Temperature Range	5% DB	MDBR	3.7	3.9	5.8	7.9	8.7	8.7	9.4	8.3	7.0	5.5	3.9	3.7
		MCDBR	3.8	4.8	7.9	11.5	11.5	12.2	12.7	10.6	9.5	6.3	4.3	4.0
		MCWBR	3.7	4.0	5.6	6.8	6.8	6.8	5.8	4.8	5.3	4.6	4.0	3.9
	5% WB	MCDBR	3.9	4.8	7.5	10.0	10.2	10.9	11.4	9.5	8.7	6.0	4.2	4.1
		MCWBR	3.8	4.1	5.6	6.4	6.6	6.8	6.0	5.0	5.3	4.6	3.9	4.0
Clear-Sky Solar Irradiance	taub	0.315	0.337	0.361	0.395	0.389	0.383	0.405	0.405	0.376	0.356	0.337	0.304	
	taud	2.285	2.310	2.307	2.264	2.349	2.394	2.351	2.357	2.421	2.409	2.329	2.256	
	Ebn at Noon	590	724	800	822	852	862	832	806	780	698	558	504	
	Edh at Noon	55	77	97	117	114	110	113	106	87	70	53	45	
All-Sky Solar Radiation	RadAvg	0.44	1.04	2.48	4.14	5.33	5.74	5.32	4.24	3.00	1.48	0.57	0.32	
	RadStd	0.08	0.17	0.23	0.49	0.50	0.42	0.57	0.30	0.29	0.19	0.09	0.03	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (67 neighbors)	+0.41	N/A	N/A	N/A	N/A	N/A	N/A	-120	-173	N/A

Nomenclature: See separate page